

Unimodality of independence polynomials of hub spiders: log-concavity certificates and a reduction to a single inequality

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Abstract

A conjecture of Alavi, Malde, Schwenk, and Erdős (1987) asserts that the independent-set sequence of every tree and forest is unimodal; it is problem 993 in the Erdős problems collection. We study the family of *hub spiders* $S(c_1, \dots, c_h)$: a root joined to h hubs, the i -th hub bearing c_i pendant paths of length two. This family contains the trees $T_{a,b,c}$ of Kadrawi–Levit, the first known trees whose independence sequence fails log-concavity, and for which unimodality was previously known only in special cases.

We develop a reduction of unimodality for this family to a single explicit inequality: the log-concavity in t of the coefficient flow $G(t) = [x^k] \prod_i ((1+2x)^{c_i} + tx(1+x)^{c_i})$ at band positions k . The route runs through: a quantitative ultra-log-concavity theorem for the hub factor $H_c = (1+2x)^c + x(1+x)^c$ with virtual degree $c+1 + \lceil c/30 \rceil$; an exact identity expressing the log-concavity defect of $I(S)$ through the relative-mass sequence $\varepsilon_k = (F_k - g_k)/g_k$; sharp per-class growth bounds obtained from exact window-mean identities and Jensen’s inequality; and the concavity of the flow, which controls all moments of the class-mass profile. The flow concavity is proved completely for $h = 2$, for arbitrary arms — by an elementary argument (a trinomial collapse, symmetrization, AM–GM, and Vandermonde) extended through a t -cancellation identity and a coefficientwise Schur lemma; at uniform multisets it is equivalent to a symmetric pairwise inequality (the *E-core*), which is proved in the limit $c \rightarrow \infty$ for all h and k by a five-line calculus bound and established for all integer c on a large finite range of (k, h) by an exact polynomial certification method. Consequently the unimodality of $I(S)$ is proved unconditionally for $h = 1$ (all c) and for 82 further families: 55 uniform ones including $S(4^5)$, a particularly fragile uniform hub spider found by the search, and the diagonal $T_{c,c,c}$ for $c \leq 10$; eleven mixed-arm two-hub members of the certificate grid; and sixteen three-hub adjacent-two-value families including $T_{3,4,4}$ and $T_{3,3,4}$, the first published non-log-concave trees. Two natural pairwise strengthenings of the flow concavity are shown to be false, with explicit hub-spider witnesses; the reduction is stated in the exact form the counterexamples leave standing.

We also record structural results from a systematic counterexample search: a balance invariant that measures distance from unimodality failure, an exact offset law for the location of log-concavity failures in uniform hub spiders, and the observation that the Levit–Mandrescu/Basit–Galvin decreasing zone protects this junction mechanism from becoming a unimodality failure.

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1 Introduction

For a finite graph G , write $i_k(G)$ for the number of independent sets of size k , $\alpha = \alpha(G)$ for the independence number, and

$$I(G; x) = \sum_{k=0}^{\alpha} i_k(G) x^k$$

for the independence polynomial. A sequence (a_k) is *unimodal* if $a_0 \leq \dots \leq a_m \geq \dots \geq a_\alpha$ for some m , and *log-concave* (LC) if $a_k^2 \geq a_{k-1}a_{k+1}$ for all internal k ; for positive sequences log-concavity implies unimodality. Alavi, Malde, Schwenk, and Erdős [1] proved that the independence sequence of a general graph can be arbitrarily far from unimodal and asked whether it is unimodal for every tree and forest; the question appears as problem 993 of [2].

The conjecture has stimulated a substantial literature. Levit and Mandrescu proved that for König–Egerváry graphs (hence all forests) the final third of the sequence is nonincreasing; Basit and Galvin [3] extended the decreasing zone and proved that random labelled trees are almost surely “almost unimodal”. Galvin and Hilyard [4] established unimodality for several recursively structured tree families. On the negative side, Kadrawi and Levit [5] gave infinite families extending the 26-vertex examples whose independence sequences are not log-concave. Galvin [6] later constructed families with log-concavity failures at unbounded independence number; machine search [7] produced many further examples; Li [8] proved unimodality for two infinite Kadrawi–Levit families; and Bautista-Ramos, Guillén-Galván and Gómez-Salgado [9] developed linear recurrences and further non-log-concave families. The hub-spider subfamily below isolates the junction mechanism in the Kadrawi–Levit examples and in many examples found by the search.

Definition 1.1 (Hub spiders). For $h \geq 1$ and $c_1, \dots, c_h \geq 1$, the *hub spider* $S(c_1, \dots, c_h)$ is the tree consisting of a root r , hubs $v_1, \dots, v_h$ adjacent to r , and, for each i , c_i pendant paths of length two attached to v_i . It has $n = 1 + h + 2C$ vertices and independence number $D = h + C$, where $C = \sum_i c_i$. We write $S(c^h)$ in the uniform case $c_1 = \dots = c_h = c$.

The family contains the Kadrawi–Levit trees $T_{a,b,c} = S(a, b, c)$, including the unstarred families treated by Li [8], and captures many hub-spider examples produced by the machine search [7]. By the root decomposition,

$$I(S(c_1, \dots, c_h); x) = P = F + xg, \quad F = \prod_{i=1}^h H_{c_i}, \quad g = (1 + 2x)^C, \quad (1)$$

where $H_c = (1 + 2x)^c + x(1 + x)^c$ is the independence polynomial of a hub with c pendant P_2 ’s, the two summands according to whether the hub is excluded or included.

1.1 Results

Our first group of results is unconditional.

Theorem 1.2 (Unconditional scope). *$I(S(c_1, \dots, c_h); x)$ has a unimodal coefficient sequence in the following cases:*

1. $h = 1$, all $c \geq 1$ (indeed I is log-concave);

2. the uniform families $S(c^h)$ with

h	1	2	3	4	5	6	7	8	9–10
$c \leq$	all	10	10	10	7	6	5	3	2

3. the eleven mixed-arm spiders of the certificate grid: $S(1, 8)$, $S(2, 4)$, $S(3, 4)$, $S(4, 7)$, $S(4, 9)$, $S(4, 11)$, $S(6, 8)$, $S(6, 12)$, $S(7, 8)$, $S(10, 12)$, $S(11, 12)$;

4. the sixteen three-hub adjacent-two-value spiders $S(a^p, (a+1)^{3-p})$, $1 \leq a \leq 8$, $p \in \{1, 2\}$ — including $T_{3,4,4} = S(3, 4, 4)$ and $T_{3,3,4} = S(3, 3, 4)$, the first published non-log-concave trees.

In particular $S(4^5)$, a particularly fragile uniform hub spider found by the search (Section 11), and the uniform Kadrawi–Levit diagonal $T_{c,c,c}$, $c \leq 10$, have unimodal independence sequences.

The proof of Theorem 1.2 is a composition of human-proved lemmas with machine-verified components, all executed in exact rational arithmetic and reproducible from the public scripts; the methodology is described in Section 9.

The structural heart of the paper is a reduction theorem.

Theorem 1.3 (Reduction to the flow concavity). *Fix $h \geq 2$ and $c_1, \dots, c_h$, and let $b_i = x(1+x)^{c_i}$, $E_i = (1+2x)^{c_i}$. Suppose that*

- (i) *the certificate inequalities of Section 9 hold for $(c_1, \dots, c_h)$ at every band position (verified exactly for every multiset in the grid $\mathcal{G}$ of Section 9, and conjectured in general, Conjecture 12.2); and*
- (ii) *at every position k in the band (4), the demotion flow $G(t) = [\prod_{l=1}^h (E_l + t b_l)]_k$ is log-concave in t on $[0, 2]$:*

$$G(t) G''(t) \leq G'(t)^2, \quad 0 \leq t \leq 2. \quad (2)$$

Then $I(S(c_1, \dots, c_h); x)$ is unimodal.

Expanding (2) by the product rule (Proposition 7.2), hypothesis (ii) is the aggregate inequality $\sum_{i \neq j} (G y_{ij} - x_i x_j) \leq \sum_i x_i^2$, where $A_l = E_l + t b_l$, $x_i = [b_i \prod_{l \neq i} A_l]_k$ and $y_{ij} = [b_i b_j \prod_{l \neq i, j} A_l]_k$. At uniform multisets the diagonal flow makes all x_i equal and all y_{ij} equal, and (ii) becomes the *symmetric E-core*

$$[b_i b_j W]_k [A_i A_j W]_k \leq \left(1 + \frac{1}{h-1}\right) [b_i A_j W]_k [b_j A_i W]_k, \quad W = \prod_{l \neq i, j} A_l, \quad (3)$$

which is the form all certifications below address.

We caution — twice — against strengthening (ii) to a *pairwise* statement for general multisets; both natural pairwise forms were intermediate conjectures of this project, refuted by adversarial scanning. The symmetric form (3) fails at lopsided pairs: at $(1, c, 1^{h-2})$ with all- E background it fails in band from $c \approx 16$, the normalized defect reaching 4.35 at $(h, c) = (8, 64)$. The weighted form $G y_{ij} - x_i x_j \leq (x_i^2 + x_j^2)/(2(h-1))$ — the exact-allocation split, which restricts to (3) at $x_i = x_j$ — fails at 1-heavy backgrounds: its smallest violation is the 107-vertex spider $S(1, 27, 1^{16})$ at $k = 5$, and at $(128, 128, 1^{192})$, $k = 7$ it fails by the factor 2.53. The failure mode is an allocation artifact: the uniform $1/(h-1)$ budget starves the heavy pair while the many

light pairs underdraw. The aggregate (ii) itself passes every exact scan — 14,410 evaluations across the flow parameter, including all 200 pairwise-violation witnesses, with worst ratio 0.085 and margins that *grow* (to -5.3 at background size 1024) exactly where the pairwise splits die (Section 9).

Four results establish (2) unconditionally on large families.

Theorem 1.4 (E-core at $h = 2$). *For $h = 2$ the symmetric form holds with constant 2 for all $c \geq 1$ and all $k \geq 0$, with no band restriction; by AM–GM this implies the flow concavity (2) at $h = 2$, $c_1 = c_2$.*

The proof (Section 8) is elementary and short: the bracket $[bE]_k$ collapses to a coefficient of the trinomial power $(1 + 3x + 2x^2)^c$, and a symmetrization, the AM–GM inequality, the Vandermonde identity, and log-concavity of binomial coefficients finish in four lines.

Theorem 1.5 (Flow concavity at $h = 2$, arbitrary arms). *For $h = 2$ with any arms (c_1, c_2) , the flow concavity (2) holds at every position $k \geq 0$, with no band restriction. In particular hypothesis (ii) of Theorem 1.3 holds for every two-hub spider.*

The proof (Section 8) combines two new lemmas with the core of Theorem 1.4: a *t-cancellation* lemma — for any fixed background the pairwise defect $Gy_{ij} - x_i x_j$ is independent of the flow parameter on the two pair slots, a unipotent congruence of the 2×2 bracket matrix — and a coefficientwise Schur lemma: among the splits $c_1 + c_2 = s$, the coefficients of $(1 + x)^{c_1} (1 + 2x)^{c_2} + (1 + x)^{c_2} (1 + 2x)^{c_1}$ are minimized at the balanced one. For even s the balanced split is the core of Theorem 1.4; odd s needs one further inequality (the *odd core*, Lemma 8.5), whose four-line proof runs on Pascal’s rule, log-concavity of a binomial row, and the numerical accident $\frac{3}{2} > \sqrt{2}$.

Theorem 1.6 (Limit calculus). *For each fixed k and $h \geq 3$, the defect of the symmetric form in the uniform case with all- E slots and background (the binding configuration in all sweeps) converges as $c \rightarrow \infty$ to*

$$\Delta_\infty(k, \ell) = \left(1 - \frac{1}{k}\right) \left(1 + \frac{1}{2\ell+2}\right)^2 \left(1 - \frac{1}{(2\ell+3)^2}\right)^k - 1, \quad \ell = h - 2,$$

and $(h - 1) \sup_{k \geq 2} \Delta_\infty(k, h - 2) \leq 1.1/(2h - 1) \leq 0.22$ for every $h \geq 3$: the limit is uniformly at least 4.5 times below the required bound.

Theorem 1.7 (Finite certification). *The symmetric form (3) (hence, at uniform multisets, the flow concavity (2)) holds, for all integer $c \geq 1$ in the uniform case, at every (k, h) with $3 \leq h \leq 7$, $k \leq 28$ and $8 \leq h \leq 10$, $k \leq 20$, and every background pattern: 996 instances, each certified by an exact polynomial argument (Section 9).*

Finally, the counterexample search that initiated this work produced extremal information of independent interest (Section 11): an exact *offset law* — in $S(c^h)$ with $c \geq 4$ the unique log-concavity failure sits exactly $h - 2$ positions below the top degree — and the *record fragility* tree, a 48-vertex junction tree (four hub gadgets and one exotic gadget) whose log-concavity *balance* (Definition 11.1) is 0.0451, more than twice the previous literature record, yet still 22 times short of what a unimodality counterexample requires.

1.2 Method overview

Write $P = F + xg$ as in (1) and let

$$\varepsilon_k = \frac{F_k - g_k}{g_k}, \quad d_k = \varepsilon_{k+1} - \varepsilon_k, \\ \rho_k = \frac{g_k}{g_{k-1}} = \frac{2(C - k + 1)}{k}, \quad \lambda_k = \frac{(k - 1)(C - k)}{(k + 1)(C - k + 2)}.$$

The engine of the reduction is the exact identity (Lemma 5.1)

$$\underbrace{2F_k g_{k-1} - F_{k-1} g_k - F_{k+1} g_{k-2}}_{\text{defect}(k)} = g_k g_{k-1} \left[(1 - \lambda_k)(1 + \varepsilon_k) + d_{k-1} - \lambda_k d_k \right],$$

which converts band log-concavity of P into a one-dimensional statement about the sequence (ε_k) , absorbed against the Newton-strength margin $\text{LC}_F(k) \geq F_k^2(N_F + 1)/((k + 1)(N_F - k + 1))$ provided by the virtual ultra-log-concavity of the factors (Theorem 4.2). Class decomposition of $F - g$, exact window-mean identities, and two Jensen arguments give closed-form two-sided bounds on every class growth; the *demotion flow* $\phi(t) = \ln[x^k] \prod_i (E_i + t b_i)$ supplies all moment bounds for the class-mass profile via its concavity — which is exactly the single inequality (2) of Theorem 1.3, and which at uniform multisets is equivalent to the symmetric E-core (3). The final assembly into explicit certificates, verified in exact rational arithmetic, is described in Section 9.

2 Setup and notation

Throughout, $[u]_k$ (or u_k , or $u[k]$) denotes the coefficient of x^k in u ; $\text{LC}_u(k) = u_k^2 - u_{k-1}u_{k+1}$; $r^u(j) = u_j/u_{j-1}$. Fixed symbols: $C = \sum_i c_i$, $D = h + C$, $n = 1 + h + 2C$, $g = (1 + 2x)^C$, $\rho_k = 2(C - k + 1)/k$, $\lambda_k = (k - 1)(C - k)/((k + 1)(C - k + 2))$, $\varepsilon_k = (F_k - g_k)/g_k$, $d_k = \varepsilon_{k+1} - \varepsilon_k$, $N_c = c + 1 + \lceil c/30 \rceil$, $N_F = \sum_i N_{c_i}$, $m_k = (N_F + 1)/((k + 1)(N_F - k + 1))$, $q_k = k/(2(C - k + 1))$, $\sigma_k = \rho_k/\rho_{k+1} - 1$.

For (1) write $b_i = x(1 + x)^{c_i}$, so $H_{c_i} = E_i + b_i$. Expanding the product over the choice of summand in each factor,

$$F - g = R = \sum_{\emptyset \neq S \subseteq [h]} t_S, \quad t_S = x^{|S|} \prod_{i \in S} (1 + x)^{c_i} \prod_{i \notin S} (1 + 2x)^{c_i},$$

and we write $W_s(k) = (\sum_{|S|=s} t_S[k])/g_k$ for the class masses, $\varepsilon_k = \sum_s W_s(k)$.

The *band* is

$$2 \leq k \leq k_A := \min(\ell_{\text{BG}} - 1, k_0 - 1, C - 1), \quad (4)$$

where $k_0 = \lceil (2D - 1)/3 \rceil$ (Levit–Mandrescu) and $\ell_{\text{BG}} = \lceil D(n - 1)/(D + n) \rceil$ (Basit–Galvin), and set $k_{\text{dec}} = \min(k_0, \ell_{\text{BG}})$. Beyond k_{dec} the sequence is nonincreasing by those theorems, and positions between C and the decreasing zone are handled separately (Section 10).

3 The composition skeleton

Lemma 3.1 (Seam). *Let P be a positive sequence of degree D , log-concave at every $k \in [1, k_{\text{dec}} - 1]$, and nonincreasing on $[k_{\text{dec}}, D]$. Then P is unimodal.*

Proof. Log-concavity at k means $r_k \geq r_{k+1}$ for $r_k = P_k/P_{k-1}$, so $r_1 \geq \dots \geq r_{k_{\text{dec}}}$ and the ratio sequence crosses 1 at most once on that range; hence P is unimodal on $[0, k_{\text{dec}}]$, with mode m say. On $[m, k_{\text{dec}}]$ the ratios are ≤ 1 , so P is nonincreasing there, and it continues nonincreasing on $[k_{\text{dec}}, D]$ by hypothesis. $\square$

The case $k = 1$ of the log-concavity hypothesis is universal: for any tree on n vertices, $i_1 = n$ and $i_2 = \binom{n}{2} - (n-1)$, and $n^2 \geq \binom{n}{2} - (n-1)$ reads $n^2 + 3n - 2 \geq 0$.

3.1 Band B

When the multiset is leaf-heavy (roughly $h \gtrsim 0.62C$) the decreasing zone begins above C , leaving positions $k \in [C, k_{\text{dec}} - 1]$ outside the band (4). These are handled directly. Since xg has support $[1, C+1]$, we have $P_k = F_k$ for $k \geq C+2$; hence for $k \geq C+3$ all three coefficients in $\text{LC}_P(k)$ are pure F and $\text{LC}_P(k) = \text{LC}_F(k) \geq 0$ by Lemma 4.1 and Lemma 4.4. At the three remaining positions $k \in \{C, C+1, C+2\}$ the corrections to $\text{LC}_F(k)$ are explicit in the top three coefficients of g :

$$\text{LC}_P(C+2) - \text{LC}_F(C+2) = -g_C F_{C+3}, \quad \text{LC}_P(C+1) - \text{LC}_F(C+1) = 2F_{C+1}g_C + g_C^2 - g_{C-1}F_{C+2},$$

and similarly at $k = C$ with g_{C-2}, g_{C-1}, g_C . The negative terms are absorbed by the margin of Theorem 4.2 using the explicit mass bound

$$\varepsilon_C \geq W_1(C) = \sum_{i=1}^h \frac{C + c_i}{2^{c_i+1}}, \quad (5)$$

which follows from the top two coefficients of $u_i = (1+x)^{c_i}(1+2x)^{C-c_i}$, namely $u_i[C] = 2^{C-c_i}$ and $u_i[C-1] = 2^{C-c_i-1}(C+c_i)$. In the leaf-heavy regime (5) is large (each $c_i \leq 2$ term contributes at least $(C+1)/8$), so the g -tail is negligible against F .

The explicit threshold is $C \geq 55$. Put $A_c = 2^c + c$, the one-below-top coefficient of H_c . Since Band B is nonempty only when $h^2 + Ch - C^2 - C > 0$, we have $h \geq 3C/5$, and if r is the number of indices with $c_i = 1$, then $r \geq 2h - C$. The reversed top expansion of F gives

$$F_C \geq e_h(A_{c_i}), \quad F_{C+1} \geq e_{h-1}(A_{c_i}), \quad F_{C+2} \geq e_{h-2}(A_{c_i}).$$

The compression inequality

$$(2^a + a)(2^b + b) \geq 3(2^{a+b-1} + a + b - 1), \quad a, b \geq 1,$$

therefore gives $F_C \geq 3^{h-1}(2^{C-h+1} + C - h + 1)$; and $A_1 = 3$, $A_c \geq 2^c$ give

$$F_{C+1} \geq r3^{r-1}2^{C-r}, \quad F_{C+2} \geq \binom{r}{2}3^{r-2}2^{C-r}.$$

Together with $N_F \leq 2C + h$ and $r^F(k+1) \leq (2C + h)(N_F - k)/((k+1)N_F)$ from Theorem 4.2, these bounds make the three absorptions at $k = C, C+1, C+2$ explicit rational-exponential inequalities. For $C \geq 55$ a quotient check shows that the five residuals are minimized on the Band B range at $(C, h) = (55, 35)$, where all are positive. The remaining finite list is exactly the leaf-heavy multisets with $C \leq 54$; these are the 24,394 integer partitions of $C - h$ over all admissible (C, h) , comprising 54,049 Band B coefficient positions. The verifier checks each position in exact integer arithmetic and finds zero failures.

4 Log-concavity of the factors

Lemma 4.1 (A0). *For every $c \geq 1$ and $t \in [0, 2]$ the polynomial $E_c + t b_c$ has a log-concave coefficient sequence. Quantitatively, for $H_c = E_c + b_c$,*

$$H_k^2 - H_{k-1}H_{k+1} \geq \frac{4^k \binom{c}{k}^2 (c+1)}{(k+1)(c-k+1)} - 2^{k-1} \binom{c}{k} \binom{c}{k-1} \geq 0.$$

Proof. With $A_k = \binom{c}{k} 2^k$ and $B_k = \binom{c}{k-1}$ decompose $\Delta_k = (A + tB)_k^2 - (A + tB)_{k-1}(A + tB)_{k+1}$ into the two pure margins and a cross term:

$$\Delta_k = \underbrace{\frac{4^k \binom{c}{k}^2 (c+1)}{(k+1)(c-k+1)}}_{A\text{-margin}} + t^2 \underbrace{\frac{\binom{c}{k-1}^2 (c+1)}{k(c-k+2)}}_{B\text{-margin}} + t 2^{k-1} \left[3 \binom{c}{k} \binom{c}{k-1} - 4 \binom{c}{k+1} \binom{c}{k-2} \right].$$

Both margins are exact evaluations of binomial log-concavity. Since $\binom{c}{k+1} \binom{c}{k-2} < \binom{c}{k} \binom{c}{k-1}$ always, the cross deficit is at most $t 2^{k-1} \binom{c}{k} \binom{c}{k-1}$, and for $t \leq 2$ the A -margin absorbs it because $2^k(c+1) \geq k(k+1)$, which holds as $2^k \geq k$ and $c+1 \geq k+1$. $\square$

Theorem 4.2 (A2: virtual ultra-log-concavity). *H_c is N_c -ULC with $N_c = c+1 + \lceil c/30 \rceil$: the sequence $H_k / \binom{N_c}{k}$ is log-concave. Consequently, by Liggett's convolution theorem [12], $F = \prod_i H_{c_i}$ is N_F -ULC with $N_F = \sum_i N_{c_i}$, and for all k*

$$\text{LC}_F(k) = F_k^2 - F_{k-1}F_{k+1} \geq F_k^2 \frac{N_F + 1}{(k+1)(N_F - k + 1)}.$$

Remark 4.3. H_c is not real-rooted for $c \geq 3$ and not plainly ULC for $c \geq 41$, so a virtual degree is necessary; N_c is nearly optimal (empirically the minimal N is $c+1$ for $c \leq 40$ and $\sim 1.032c$ asymptotically).

Proof. Write $C_j = \binom{c}{j}$, $A_k = C_k 2^k$, $B_k = C_{k-1}$, $N = N_c$, and $\Delta = H_k^2 k(N-k) - H_{k-1}H_{k+1}(k+1)(N-k+1)$; we must show $\Delta \geq 0$ for $1 \leq k \leq c$. Three exact evaluations, using $C_{k-1}C_{k+1}/C_k^2 = k(c-k)/((k+1)(c-k+1))$ and the telescoping $(N-k)(c-k+1) - (N-k+1)(c-k) = N-c$:

$$\begin{aligned} \text{A-part: } & A_k^2 k(N-k) - A_{k-1}A_{k+1}(k+1)(N-k+1) = 2^k C_k C_{k-1} \cdot 2^k (N-c); \\ \text{B-part: } & B_k^2 k(N-k) - B_{k-1}B_{k+1}(k+1)(N-k+1) = \frac{C_{k-1}^2}{k(c-k+2)} \left[k^2(N-c-1) \right. \\ & \quad \left. + (c-k+1)(N-k+1) \right] \geq 0; \\ \text{cross: } & 2A_k B_k k(N-k) - (A_{k-1}B_{k+1} + A_{k+1}B_{k-1})(k+1)(N-k+1) \\ & = 2^k C_k C_{k-1} \left[2k(N-k) - (k+1)(N-k+1) \left(\frac{1}{2} + 2\theta_k \right) \right], \\ & \theta_k = \frac{(k-1)(c-k)}{(k+1)(c-k+2)}. \end{aligned}$$

Dropping the nonnegative B -part and clearing denominators, it suffices that

$$S(k, c, N) := 2(c-k+2)[2^k(N-c) + 2k(N-k)] - (N-k+1)[(k+1)(c-k+2) + 4(k-1)(c-k)] \geq 0.$$

S is nondecreasing in N (the N -derivative is $2(c-k+2)(2^k+2k) - [(k+1)(c-k+2) + 4(k-1)(c-k)] \geq 0$ since $2^{k+1} \geq k+1$ and $4k(c-k+2) \geq 4(k-1)(c-k)$), so it is enough to prove $S \geq 0$ at the real value $N = c+1+c/30$. For $k \leq 12$, $P_k(c) := 30 S(k, c, c+1+c/30)$ is a quadratic in c with positive leading coefficient and $P_k(k) \geq 0$, $P'_k(k) \geq 0$ — for example $P_1 = 6(c+1)(11c+10)$, $P_2 = c(39c+248)$, ..., $P_{12} = 7913c^2 + 170618c - 2496600$ — hence $P_k \geq 0$ on $c \geq k$. For $k \geq 13$, use $(k+1)(c-k+2) + 4(k-1)(c-k) \leq 5(k+1)(c-k+2)$, $N-k+1 \leq N+1 \leq \frac{31}{30}(c+30)$, and $N-c \geq \frac{1}{30}(c+30)$:

$$S \geq \frac{(c-k+2)(c+30)}{30} [2^{k+1} - 155(k+1)] \geq 0,$$

since $2^{k+1} \geq 155(k+1)$ for $k \geq 13$. (The inequality $S \geq 0$ was additionally verified exactly for every $c \leq 3000$ and every $k \leq c$.) The Liggett corollary follows since zero-padding a degree- d N -ULC sequence to virtual degree N preserves log-concavity of the normalized sequence. $\square$

Lemma 4.4 (Closure). *The convolution of positive log-concave sequences with no internal zeros is log-concave. Quantitatively, $\text{LC}_{u*v}(k) \geq \sum_j \text{LC}_u(k-1-j) \text{LC}_v(j+1)$.*

Proof sketch. The qualitative statement is classical (positive LC sequences with no internal zeros are Pólya frequency of order 2, and PF_2 is closed under convolution; see e.g. [10]). The quantitative form follows from the Cauchy–Binet expansion of the 2×2 minors of the convolution into sums of products of paired spread minors of u and v , of which the displayed sum keeps the adjacent pairs; it is stated for completeness but only the qualitative statement is used in this paper (the quantitative margins come from Theorem 4.2). $\square$

Lemma 4.5 (D1: the case $h=1$). *For every $c \geq 1$, $I(S(c)) = H_c + xE_c$ is log-concave; in fact the mixed defect $2H_k g_{k-1} - H_{k-1} g_k - H_{k+1} g_{k-2}$ is nonnegative for all $1 \leq k \leq c$, where $g = E_c$.*

Proof. Split $H = g + b$, $b_k = \binom{c}{k-1}$; the defect is linear in its first argument, defect = $W(g, g) + W(b, g)$. The pure part has the exact closed form $W(g, g)(k) = g_k g_{k-1} - g_{k+1} g_{k-2} = g_k g_{k-1} (2c+2) / ((k+1)(c-k+2)) > 0$, and the mixed part evaluates to $W(b, g)(k) = 2^{k-2} [4 \binom{c}{k-1}^2 - 5 \binom{c}{k-2} \binom{c}{k}] \geq -2^{k-2} \binom{c}{k-1}^2$ by log-concavity of the binomial row. Absorption: $g_k g_{k-1} = \binom{c}{k} \binom{c}{k-1} 2^{2k-1}$, so $W(g, g) \geq 2^{k-2} \binom{c}{k-1}^2$ reduces to $2^{k+2}(c+1)(c-k+1) \geq k(k+1)(c-k+2)$, which holds since $c-k+1 \geq (c-k+2)/2$ and $2^{k+1}(c+1) \geq k(k+1)$. Log-concavity of $I(S(c))$ then follows from Lemma 4.1, log-concavity of the shifted binomial xg , and $\text{LC}_P = \text{LC}_H + \text{LC}_{xg} + \text{defect}$. $\square$

5 The ε -calculus and the master inequality

Lemma 5.1 (The ε -identity). *With the notation of Section 1.2,*

$$2F_k g_{k-1} - F_{k-1} g_k - F_{k+1} g_{k-2} = g_k g_{k-1} [(1 - \lambda_k)(1 + \varepsilon_k) + d_{k-1} - \lambda_k d_k].$$

Moreover $\text{LC}_P(k) \geq \text{defect}(k) + \text{LC}_F(k)$, and with the margin of Theorem 4.2 the band log-concavity of P follows from the master inequality

$$\lambda_k d_k - d_{k-1} \leq (1 - \lambda_k)(1 + \varepsilon_k) + \rho_k m_k (1 + \varepsilon_k)^2, \quad m_k = \frac{N_F + 1}{(k+1)(N_F - k + 1)}. \quad (6)$$

Proof. Substituting $F_j = (1 + \varepsilon_j)g_j$ and dividing by $g_k g_{k-1}$, the left side becomes

$$2(1 + \varepsilon_k) - (1 + \varepsilon_{k-1}) - (1 + \varepsilon_{k+1}) \frac{g_{k+1}g_{k-2}}{g_k g_{k-1}} = 2(1 + \varepsilon_k) - (1 + \varepsilon_{k-1}) - (1 + \varepsilon_{k+1}) \frac{\rho_{k+1}}{\rho_{k-1}},$$

and $\rho_{k+1}/\rho_{k-1} = \lambda_k$ by the closed form of ρ . Collecting terms gives the identity. For the reduction, write $\text{LC}_P(k) = \text{LC}_F(k) + \text{LC}_{xg}(k) + \text{defect}(k)$ with $\text{defect}(k) = 2F_k g_{k-1} - F_{k-1} g_k - F_{k+1} g_{k-2}$ (the mixed terms of the square minus the two mixed products); $\text{LC}_{xg}(k) \geq 0$ because shifted binomial sequences are log-concave. If (6) holds then by the identity

$$\text{defect}(k) \geq -g_k g_{k-1} \rho_k m_k (1 + \varepsilon_k)^2 = -m_k g_k^2 (1 + \varepsilon_k)^2 = -m_k F_k^2 \geq -\text{LC}_F(k)$$

by Theorem 4.2, using $g_k g_{k-1} \rho_k = g_k^2$ and $F_k = (1 + \varepsilon_k)g_k$. Hence $\text{LC}_P(k) \geq 0$. $\square$

6 Per-class growth bounds

Group the classes by (s, a) with $s = |S|$ and $a = \sum_{i \in S} c_i$: the class mass is $w_{s,a}(k) = \nu_{s,a} u[k - s]/g_k$ with $u = (1 + x)^a (1 + 2x)^{C-a}$ and $\nu_{s,a}$ the number of subsets realizing (s, a) , so that $W_s(k) = \sum_a w_{s,a}(k)$. All bounds below concern the *growth* $\gamma(k) = w(k)/w(k-1) = r^u(k-s)/\rho_k$, where $r^u(j) = u_j/u_{j-1}$.

Lemma 6.1 (E1: ratio sandwich). *For $u = (1 + x)^a (1 + 2x)^{C-a}$ and all j in the support,*

$$\frac{C - j + 1}{j} \leq r^u(j) \leq \frac{2(C - j + 1)}{j}.$$

Proof. $(1 + x) \leq_{\text{lr}} (1 + 2x)$ (the coefficient ratio $1 : 2$ exceeds $1 : 1$), and likelihood-ratio order between nonnegative sequences is preserved by convolution with a common PF_2 (log-concave, no internal zeros) factor [11, Thm. 1.C.9]; equivalently, a two-line Cauchy–Binet check. Applying this a resp. $C - a$ times, $(1 + x)^C \leq_{\text{lr}} u \leq_{\text{lr}} (1 + 2x)^C$, and $X \leq_{\text{lr}} Y$ implies $r^X(j) \leq r^Y(j)$ wherever defined; the two outer ratios are $(C - j + 1)/j$ and $2(C - j + 1)/j$. $\square$

Two corollaries used throughout: $\varepsilon_{k-1} \leq 2\varepsilon_k$ on the band, and $W_s(k) \leq \binom{h}{s} q_k^s$ with $q_k = k/(2(C - k + 1))$.

Lemma 6.2 (Exact window means). *Write $u = (1 + x)\tilde{u}$ with $\tilde{u} = (1 + x)^{a-1} (1 + 2x)^{C-a}$, and let I (resp. I') be the number of $(1 + x)$ -factors used at level $j - 1$ (resp. j) under the coefficient measure of u . Then*

$$\mathbb{E}[I] = \frac{a}{1 + r^{\tilde{u}}(j-1)}, \quad \mathbb{E}[I'] = \frac{a}{1 + r^{\tilde{u}}(j)},$$

and consequently, by Lemma 6.1 applied to $\tilde{u}$,

$$\mathbb{E}[I] \geq \frac{a(j-1)}{(j-1) + 2(C-j+1)}, \quad \mathbb{E}[I'] \leq \frac{aj}{C}.$$

Proof. $\sum_i i \binom{a}{i} v_{j-1-i} = a \sum_i \binom{a-1}{i-1} v_{j-1-i} = a \tilde{u}_{j-2}$ for $v = (1 + 2x)^{C-a}$, while $u_{j-1} = \tilde{u}_{j-1} + \tilde{u}_{j-2}$; divide. The second identity is the same computation one level up. $\square$

Lemma 6.3 (Jensen floor and ceiling). *With $r^v(m) = 2(C - a - m + 1)/m$,*

$$r^u(j) \geq r^v(j - \mathbb{E}[I]) \quad \text{and} \quad r^u(j) \leq r^v(j - \mathbb{E}[I']).$$

Combined with Lemma 6.2 these give closed-form rational bounds $\text{fl}(s, a, k) \leq \gamma(k) \leq \text{ce}(s, a, k)$ for every class growth.

Proof. Conditioning on I , $r^u(j) = \mathbb{E}_\omega[r^v(j - I)]$ where ω is the level- $(j - 1)$ window measure; $i \mapsto r^v(j - i)$ is convex and increasing, so Jensen plus the floor on $\mathbb{E}[I]$ give the first bound. For the second, $1/r^u(j) = \mathbb{E}_{\omega'}[1/r^v(j - I')]$ under the level- j window measure, and $i \mapsto 1/r^v(j - i)$ is convex and decreasing; Jensen plus the ceiling on $\mathbb{E}[I']$ finish. $\square$

The charge decomposition. The left side of (6) splits exactly over classes:

$$\lambda_k d_k - d_{k-1} = \sum_{s,a} w_{s,a}(k) \left[\lambda_k (\gamma_+ - 1) - \left(1 - \frac{1}{\gamma_0}\right) \right],$$

where γ_0, γ_+ are the class growths into k and $k + 1$. Each bracket admits two upper bounds (“strategies”):

$$\text{cap}_A = \lambda_k (\text{ce}_+ - 1)^+ + \frac{(1 - \text{fl})^+}{\text{fl}}, \quad \text{cap}_B = \lambda_k \left(\text{ce}_0 \sigma_k + \frac{\text{dev}^2}{\text{fl}} \right) + (1 - \lambda_k) \frac{(1 - \text{fl})^+}{\text{fl}},$$

with $\sigma_k = \rho_k / \rho_{k+1} - 1$ and $\text{dev} = \max(\text{ce}_0 - 1, 1 - \text{fl})$; strategy B uses the second-difference identity $\gamma_+ - 2 + 1/\gamma_0 = (\gamma_+ - \gamma_0) + (\gamma_0 - 1)^2 / \gamma_0$ together with $\gamma_+ \leq \gamma_0(1 + \sigma_k)$ (log-concavity of u). Since both are upper bounds for the *same* bracket, the per-class minimum $\min(\text{cap}_A, \text{cap}_B)$ is a valid charge, and

$$\lambda_k d_k - d_{k-1} \leq \sum_s W_s(k) \text{cap}(s), \quad \text{cap}(s) = \max_a \min(\text{cap}_A, \text{cap}_B). \quad (7)$$

7 The demotion flow

Fix a band position k and define the *demotion flow*

$$G(t) = \left[\prod_{i=1}^h (E_i + t b_i) \right]_k, \quad \phi(t) = \ln G(t), \quad t \in [0, 2].$$

Then $G(1) = F_k$, $G(0) = g_k$, and expanding the product over which factors contribute b_i ,

$$G(t) = g_k \sum_{s \geq 0} W_s(k) t^s,$$

so the flow is the generating function of the class-mass profile. Three exact facts:

$$\phi(1) - \phi(0) = \ln(1 + \varepsilon_k), \quad \phi'(1) = \sum_{i=1}^h \frac{\delta_i}{1 + \delta_i}, \quad \sum_s s W_s(k) = (1 + \varepsilon_k) \phi'(1),$$

where $\delta_i = (F - F^{[i]})_k / F_k^{[i]}$ and $F^{[i]}$ is F with hub i demoted ($H_i \rightarrow E_i$); the second uses $b_i \prod_{l \neq i} H_l = F - F^{[i]}$.

Proposition 7.1 (Flow facts). *Suppose $\phi'' \leq 0$ on $[0, 2]$ at position k . Then*

$$\begin{aligned} \text{(P5)} \quad & \sum_s s W_s \leq (1 + \varepsilon) \ln(1 + \varepsilon), \\ \text{(m2)} \quad & \sum_s s(s - 1) W_s \leq (1 + \varepsilon) \ln^2(1 + \varepsilon), \\ \text{(envelope +)} \quad & \sum_{s \geq 1} W_s t^s \leq (1 + \varepsilon)^t - 1 \quad (1 \leq t \leq 2), \\ \text{(envelope -)} \quad & \sum_{s \geq 1} W_s t^s \geq (1 + \varepsilon)^t - 1 \quad (0 \leq t \leq 1). \end{aligned}$$

Proof. Concavity gives $\phi'(1) \leq \phi(1) - \phi(0) = \ln(1 + \varepsilon)$, which is (P5) after multiplying by $1 + \varepsilon$; $G''G \leq (G')^2$ at $t = 1$ gives $\sum_s s(s-1)W_s \leq (1 + \varepsilon)\phi'(1)^2$, which is (m2) by (P5). For the envelopes: on $[1, 2]$, $\phi(t) \leq \phi(1) + (t-1)\phi'(1) \leq \ln F_k + (t-1)\ln(1 + \varepsilon)$ (tangent line plus (P5)), and on $[0, 1]$, $\phi(t) \geq (1-t)\phi(0) + t\phi(1)$ (chord); exponentiate and divide by g_k . $\square$

Proposition 7.2 (Diagonal dominance). *With $A_l = E_l + tb_l$, $x_i = [b_i \prod_{l \neq i} A_l]_k$ and $y_{ij} = [b_i b_j \prod_{l \neq i, j} A_l]_k$,*

$$(G')^2 - GG'' = \left(\sum_i x_i \right)^2 - G \sum_{i \neq j} y_{ij} = \sum_i x_i^2 + \sum_{i \neq j} (x_i x_j - G y_{ij}),$$

so hypothesis (ii) of Theorem 1.3 is the aggregate inequality $\sum_{i \neq j} (G y_{ij} - x_i x_j) \leq \sum_i x_i^2$, and $\phi''(t) \leq 0$ holds at any (k, t) where all the x_i are equal and the symmetric form (3) holds for every pair: there each ordered pair contributes at most $x^2/(h-1)$, and the $h(h-1)$ ordered pairs sum to exactly $\sum_i x_i^2$.

Proof. The identity is the product rule for $G' = \sum_i x_i$ -type derivatives of the multiaffine flow; the symmetric case is the stated count. $\square$

Remark 7.3 (Pairwise splits fail). The natural per-pair sufficient conditions for the aggregate are false in general. Allocating the budget $\sum_i x_i^2$ uniformly, $G y_{ij} - x_i x_j \leq (x_i^2 + x_j^2)/(2(h-1))$ (exact at symmetric points), is violated by honest in-band instances at 1-heavy backgrounds — smallest witness $S(1, 27, 1^{16})$, $n = 107$, at $k = 5$; worst observed factor 7.5 at $(1, 128, 1^{256})$, $k = 10$ — while the aggregate holds there with the margin growing without bound. Any pairwise route to hypothesis (ii) for mixed multisets must therefore allocate adaptively; the aggregate is the invariant object.

There is, however, a sufficient route that eliminates the flow parameter entirely. Write $G(t) = \sum_s W_s(k) t^s$, where

$$W_s(k) = \sum_{|S|=s} \left[\prod_{i \in S} b_i \prod_{i \notin S} E_i \right]_k$$

is the class-mass profile (contiguous support $[0, \min(h, k)]$ at band positions).

Proposition 7.4 (Factorial criterion). *Let $G(t) = \sum_s W_s t^s$ with $W_s \geq 0$ and contiguous support. If the sequence $(s! W_s)$ is log-concave on the support, then G is log-concave on $(0, \infty)$. Consequently hypothesis (ii) at position k follows from the ladder*

$$s W_s(k)^2 \geq (s+1) W_{s-1}(k) W_{s+1}(k), \quad 1 \leq s \leq \min(h, k) - 1. \quad (8)$$

Proof. Log-concavity of $(s! W_s)$ says $\rho_s := s W_s / W_{s-1}$ is nonincreasing on the support. Under the tilted law $\mathbb{P}_t(S = s) = W_s t^s / G(t)$,

$$(\ln G)'(t) = \frac{G'(t)}{G(t)} = \sum_s \frac{(s+1) W_{s+1} t^s}{G(t)} = \mathbb{E}_t[\rho_{S+1}]$$

(with $\rho := 0$ above the support). For $t' > t$ the likelihood ratio $\mathbb{P}_{t'}(s) / \mathbb{P}_t(s) \propto (t'/t)^s$ is increasing in s , so $\mathbb{P}_{t'}$ dominates $\mathbb{P}_t$ in the likelihood-ratio order, hence stochastically [11]; since ρ_{S+1} is nonincreasing, $\mathbb{E}_t[\rho_{S+1}]$ is nonincreasing in t . Thus $(\ln G)'$ is nonincreasing on $(0, \infty)$. $\square$

The ladder (8) is tight at its base and unifies the results of this paper: the rung $s = 1$ reads $W_1^2 \geq 2W_0W_2$, which is *identically* hypothesis (ii) at $t = 0$ (so the bottom rung is necessary); at $h = 2$ the ladder is that single rung, which is the inequality proved in Theorem 1.5; and at uniform multisets, where $W_s = \binom{h}{s} u_s$ with $u_s = [b^s E^{h-s}]_k$, rung s becomes $u_s^2 \geq u_{s-1}u_{s+1}(h-s)/(h-s+1)$, whose first case is the symmetric E-core (3). Each bracket u_s is a polynomial in c at fixed (k, h, s) , so the higher rungs are certifiable by the method of Section 9.2. Exact scans (Section 9) found no rung failure anywhere — and located the binding rung at $s = 1$ throughout — so the ladder is the route we pursue for mixed multisets (Section 12). We also record the polynomial identity behind rung 1: with $\Phi_s = \sum_{|S|=s} \prod_{i \in S} b_i \prod_{i \notin S} E_i$ (so $W_s = [\Phi_s]_k$), the cross terms in Φ_1^2 cancel against $2\Phi_0\Phi_2$ exactly, leaving

$$\Phi_1^2 - 2\Phi_0\Phi_2 = \sum_i b_i^2 \left(\prod_{l \neq i} E_l \right)^2,$$

which is coefficientwise nonnegative; rung 1 is the localization of this identity to the (k, k) product of coefficients.

The class masses also carry an exact majorization structure.

Proposition 7.5 (Schur-convexity of the class masses). *At fixed (C, h, k, s) , the class mass W_s is Schur-convex on multisets with sum C and h parts; in particular $W_s(c) \geq W_s(\text{bal}(C, h))$ for every multiset c , where $\text{bal}(C, h)$ is the balanced two-value partition.*

Proof. A Robin Hood transfer toward balance on slots (i, j) with $c_i < c_j$ induces, on the multiset of s -subset sums $\sigma_S = \sum_{l \in S} c_l$, the simultaneous transfers $(\sigma_{S \cup i}, \sigma_{S \cup j}) \rightarrow (\sigma_{S \cup i} + 1, \sigma_{S \cup j} - 1)$ over all $(s-1)$ -subsets S avoiding i, j — each toward balance, since $\sigma_S + c_i < \sigma_S + c_j$; subsets containing both or neither are unchanged. Hence the subset-sum multiset on the spread side majorizes the balanced side's. The summand $u(\sigma) = [x^{k-s}](1+x)^\sigma(1+2x)^{C-\sigma}$ is convex in σ , by the exact second difference $\Delta^2 u = [x^{k-s-2}](1+x)^\sigma(1+2x)^{C-\sigma-2} \geq 0$ (linear at $k-s=1$, constant at $k-s=0$), and Karamata's inequality concludes. $\square$

Thus both sides of every rung grow as the multiset spreads, and the open content of the ladder for mixed multisets is a relative-rate statement (the domination lemma of Section 12).

8 The E-core and the flow concavity at $h = 2$

The two reduction lemmas of this section concern the symmetric form (3); they power the uniform-case certification, where by Proposition 7.2 the symmetric form at every pair is equivalent to the flow concavity. For mixed multisets no pairwise reduction is available (Remark 7.3) and the aggregate diagonal t -family must be treated as such (Section 12); the exception is $h = 2$, where the aggregate *is* the single-pair inequality and is proved below for even total degree.

Lemma 8.1 (Vertex reduction). *The supremum over $(t_1, \dots, t_h) \in [0, 2]^h$ of the symmetric-form defect $[b_i b_j W][A_i A_j W] / ([b_i A_j W][b_j A_i W])$ with $A_l = E_l + t_l b_l$ is attained with each $A_l \in \{E_l, E_l + 2b_l\}$.*

Proof. The defect is a ratio of polynomials with positive coefficients, multiaffine in the t_l ; as a function of each t_l separately it is a Möbius transformation with no pole on $[0, \infty)$, hence monotone there, so the supremum over the box is attained at a vertex. $\square$

Lemma 8.2 (Slot expansion). *For any nonnegative background W , any $d \geq 0$, and $H_2 := E + 2b$, the inequalities with $X_i, X_j \in \{E, H_2\}$ all follow termwise from the single core case $X_i = X_j = E$ with the same W and constant $1 + d$.*

Proof. Expand $[H_2X] = [EX] + 2[bX]$ in each slot; every cross term appears on both sides with matching or dominating coefficients; the only nontrivial comparison is the core case. $\square$

Proof of Theorem 1.4. By Lemmas 8.1 and 8.2 it suffices to prove $[b^2]_k[E^2]_k \leq 2([bE]_k)^2$. All three brackets are explicit: $[b^2]_k = \binom{2c}{k-2}$, $[E^2]_k = \binom{2c}{k}2^k$, and since $(1+x)^c(1+2x)^c = (1+3x+2x^2)^c$,

$$[bE]_k = T_{k-1}, \quad T_n := [x^n](1+3x+2x^2)^c = \sum_m \binom{c}{m} \binom{c}{n-m} 2^{n-m}.$$

By the symmetry $m \leftrightarrow n-m$ of $\binom{c}{m}\binom{c}{n-m}$ and AM-GM, followed by Vandermonde,

$$T_n = \frac{1}{2} \sum_m \binom{c}{m} \binom{c}{n-m} (2^m + 2^{n-m}) \geq 2^{n/2} \sum_m \binom{c}{m} \binom{c}{n-m} = 2^{n/2} \binom{2c}{n}.$$

Hence, using log-concavity of the binomial row $\binom{2c}{\cdot}$,

$$2T_{k-1}^2 \geq 2^k \binom{2c}{k-1}^2 \geq 2^k \binom{2c}{k-2} \binom{2c}{k} = [b^2]_k [E^2]_k. \quad \square$$

Lemma 8.3 (t -cancellation). *Fix any background polynomial W and a position k , and write $A_i = E_i + t b_i$, $A_j = E_j + t b_j$ on the two pair slots. Then*

$$[b_i b_j W]_k [A_i A_j W]_k - [b_i A_j W]_k [b_j A_i W]_k = [b_i b_j W]_k [E_i E_j W]_k - [b_i E_j W]_k [b_j E_i W]_k,$$

independent of t . Consequently, since the brackets $[b_i A_j W]_k = [b_i E_j W]_k + t[b_i b_j W]_k$ are non-decreasing in t , any upper bound on the pairwise defect that is monotone in the cross-brackets need only be verified at $t = 0$ on the pair slots.

Proof. Brackets are linear in each slot polynomial, so with $P = [b_i b_j W]_k$, $Q_0 = [E_i E_j W]_k$, $X_0 = [b_i E_j W]_k$, $Y_0 = [b_j E_i W]_k$, the four t -dependent brackets are $Q_0 + t(X_0 + Y_0) + t^2 P$, $X_0 + tP$, and $Y_0 + tP$; expanding, $P(Q_0 + t(X_0 + Y_0) + t^2 P) - (X_0 + tP)(Y_0 + tP) = P Q_0 - X_0 Y_0$. Equivalently, the matrix $M(t) = \begin{pmatrix} P & Y \\ X & Q \end{pmatrix}$ satisfies $M(t) = (I + t e_2 e_1^T) M(0) (I + t e_1 e_2^T)$, a unipotent congruence, which preserves the determinant. $\square$

Lemma 8.4 (Schur minimality of the balanced split). *For $a + b = s$ let $S(a, b) = (1+x)^a(1+2x)^b + (1+x)^b(1+2x)^a$. Then for $a < b$,*

$$S(a, b) - S(a+1, b-1) = x [(1+x)(1+2x)]^a [(1+2x)^{b-a-1} - (1+x)^{b-a-1}] \geq 0$$

coefficientwise; hence the coefficients of S are minimized, among the splits of s , at the balanced one.

Proof. With $d = b - a$, $S(a, b) = [(1+x)(1+2x)]^a T_d$ where $T_d = (1+x)^d + (1+2x)^d$, and $T_d - (1+x)(1+2x)T_{d-2} = (1+x)^{d-1}[(1+x) - (1+2x)] + (1+2x)^{d-1}[(1+2x) - (1+x)] = x[(1+2x)^{d-1} - (1+x)^{d-1}]$, whose coefficients are nonnegative because $(1+2x)^{d-1}$ dominates $(1+x)^{d-1}$ coefficientwise. $\square$

Lemma 8.5 (Odd core). *Let $s = 2m + 1$, $m \geq 1$, and $T_n = [x^n](1 + 3x + 2x^2)^m$. Then for all $k \geq 2$,*

$$2 \binom{s}{k-2} 2^k \binom{s}{k} \leq (2T_{k-1} + 3T_{k-2})^2.$$

Proof. For $k > s$ the left side vanishes; let $2 \leq k \leq s$ and put $a = \binom{2m}{k-2} \geq 1$, $b = \binom{2m}{k-1} \geq 1$. By Pascal's rule and log-concavity of the row $\binom{2m}{\cdot}$,

$$\binom{s}{k-2} = a + \binom{2m}{k-3} \leq a \left(1 + \frac{a}{b}\right), \quad \binom{s}{k} = b + \binom{2m}{k} \leq b \left(1 + \frac{b}{a}\right),$$

so $\binom{s}{k-2} \binom{s}{k} \leq (a+b)^2$. The AM–GM/Vandermonde bound from the proof of Theorem 1.4, $T_n \geq 2^{n/2} \binom{2m}{n}$, gives

$$2T_{k-1} + 3T_{k-2} \geq 2^{k/2} \left(\sqrt{2}b + \frac{3}{2}a \right) \geq 2^{k/2} \sqrt{2}(a+b),$$

the last step because $\frac{3}{2} > \sqrt{2}$. Squaring, $(2T_{k-1} + 3T_{k-2})^2 \geq 2^{k+1}(a+b)^2 \geq 2 \binom{s}{k-2} 2^k \binom{s}{k}$. $\square$

The constant 3, inherited from $T_1 = 2 + 3x$ at the balanced split, is exactly what the proof needs: a cross term βx with $\beta/2 < \sqrt{2}$ would fail. The lemma is also exactly verified for all $s \leq 161$ and all k , with worst ratio 0.5217 — the same strength as the even core (0.5218).

Proof of Theorem 1.5. At $h = 2$ there is no background and $(G')^2 - GG'' = x_1^2 + x_2^2 + 2(x_1x_2 - Gy)$ with $G = [A_1A_2]_k$, $y = [b_1b_2]_k$, $x_i = [b_iA_j]_k$, so (2) reads $2(Gy - x_1x_2) \leq x_1^2 + x_2^2$. By Lemma 8.3 (with $W = 1$) the left side is independent of t while the right side is nondecreasing in t , so it suffices to prove, at $t = 0$,

$$2[b_1b_2]_k [E_1E_2]_k \leq (X_0 + Y_0)^2, \quad X_0 = [b_1E_2]_k, \quad Y_0 = [b_2E_1]_k.$$

Here $[b_1b_2]_k = \binom{s}{k-2}$ and $[E_1E_2]_k = 2^k \binom{s}{k}$ depend only on $s = c_1 + c_2$, while $X_0 + Y_0 = [x^{k-1}]S(c_1, c_2)$ is minimized at the balanced split by Lemma 8.4. For even s , $[x^{k-1}]S(s/2, s/2) = 2[x^{k-1}](1 + 3x + 2x^2)^{s/2} = 2T_{k-1}$, and the display in the proof of Theorem 1.4, at $c = s/2$, states exactly $\binom{s}{k-2} 2^k \binom{s}{k} \leq 2T_{k-1}^2$, i.e. $2[b_1b_2]_k [E_1E_2]_k \leq (2T_{k-1})^2 \leq (X_0 + Y_0)^2$. For odd s , $S(\frac{s-1}{2}, \frac{s+1}{2}) = (1 + 3x + 2x^2)^{(s-1)/2}((1+x) + (1+2x))$, so $[x^{k-1}]S_{\text{bal}} = 2T_{k-1} + 3T_{k-2}$ with T at exponent $(s-1)/2$, and Lemma 8.5 closes the chain. $\square$

Proof of Theorem 1.6. With $A = 2\ell + 2$, $B = 2\ell + 3$, the bounds $\ln(1-x) \leq -x$ and $\ln(1+x) \leq x$ give $\ln(1 + \Delta_\infty) \leq -\frac{1}{k} + \frac{2}{A} - \frac{k}{B^2}$, and by AM–GM $\frac{1}{k} + \frac{k}{B^2} \geq \frac{2}{B}$, so $\ln(1 + \Delta_\infty) \leq 2/(AB) \leq 1/6$. Since $e^x - 1 \leq 1.1x$ on $(0, 1/6]$,

$$(h-1)\Delta_\infty \leq (\ell+1) \cdot \frac{2.2}{(2\ell+2)(2\ell+3)} = \frac{1.1}{2\ell+3} \leq 0.22. \quad \square$$

Derivation of Δ_∞ . In the uniform all- E background, fix k and let $c \rightarrow \infty$. The needed bracket types are $[b^2E^{h-2}]_k$, $[E^h]_k$, and $[bE^{h-1}]_k$. In each finite convolution sum,

$$\binom{ac}{m} 2^m = \frac{(2ac)^m}{m!} (1 + O(m^2/(ac))),$$

and therefore

$$\begin{aligned} [E^h]_k &= \binom{hc}{k} 2^k \sim \frac{(2hc)^k}{k!}, \\ [bE^{h-1}]_k &= \sum_m \binom{c}{m} \binom{(h-1)c}{k-1-m} 2^{k-1-m} \\ &\sim \frac{(2(h-1)c)^{k-1}}{(k-1)!} \sum_m \binom{k-1}{m} \left(\frac{1}{2(h-1)}\right)^m, \end{aligned}$$

and the binomial sum equals $(1 + \frac{1}{2(h-1)})^{k-1}$, giving $[bE^{h-1}]_k \sim ((2h-1)c)^{k-1}/(k-1)!$; similarly $[b^2E^{h-2}]_k \sim (2(h-1)c)^{k-2}/(k-2)!$ via $2(h-2)(1 + \frac{1}{h-2}) = 2(h-1)$. Hence, with $B = 2h-1$,

$$\Delta_\infty + 1 = \frac{(2(h-1))^{k-2}(2h)^k}{B^{2k-2}} \cdot \frac{(k-1)!^2}{(k-2)!k!} = \left(1 - \frac{1}{B^2}\right)^{k-2} \left(1 + \frac{1}{B}\right)^2 \left(1 - \frac{1}{k}\right),$$

and the identity $(1 - B^{-2})^{-2}(1 + B^{-1})^2 = (1 + \frac{1}{B-1})^2$ converts this to the form displayed in Theorem 1.6 with $B-1 = 2\ell+2$. $\square$

The finite- c statement of Theorem 1.7 is proved by the polynomial certification method of Section 9.

9 Certificates and machine verification

This section describes the machine-verified components and the protocol under which they were produced and audited. All verifications are exact: every inequality is decided in rational arithmetic (Python `fractions.Fraction` or `sympy` integer polynomials), with no floating point in any verified statement.

9.1 Certificates for the master inequality

For each multiset and band position, the master inequality (6) is established through (7) by one of two explicit certificates.

Certificate T1 (unconditional). From the identity $1 + \varepsilon_{k+1} = (r_{k+1}/\rho_{k+1})(1 + \varepsilon_k)$ and $r_{k+1} \leq r_k$ (log-concavity of F , by Lemmas 4.1 and 4.4),

$$\lambda_k(1 + \varepsilon_k) \left[(1 + \sigma_k) \frac{1 + \varepsilon_k}{1 + \varepsilon_{k-1}} - 1 \right] + (\varepsilon_{k-1} - \varepsilon_k)^+ \leq \text{RHS of (6)}$$

suffices, a single rational inequality depending only on Lemmas 4.1, 4.4 and Theorem 4.2.

Certificate T2 (uses the flow facts, hence the flow concavity (2) at the same position). A quadratic majorant $\mu + y_1s + y_2s(s-1) \geq \text{cap}(s)$, anchored at $s = 1, 2$ with y_2 the maximal second divided difference (clamped so $y_1 \geq 0$), is played against the *rationalized* moment rows

$$\sum_s sW_s \leq R_1 = \frac{\varepsilon(2 + \varepsilon)}{2}, \quad \sum_s s(s-1)W_s \leq R_2 = \frac{\varepsilon^2(2 + \varepsilon)^2}{4(1 + \varepsilon)},$$

which follow from Proposition 7.1 and the trapezoid bound $\ln(1+x) \leq x(2+x)/(2(1+x))$; the residual tail is covered by the class bounds $W_s \leq \binom{h}{s} q_k^s$ or the $t = 2$ envelope, whichever is

cheaper. Pointwise domination is checked exactly for every s , and the certificate obligation is one rational comparison.

These certificate families were located by linear programming (maximize the charge sum over all profiles (W_s) satisfying the proved constraints; the optimum is 25.97% below the requirement at the worst point of the grid), and the dual solutions were then replaced by the closed-form constructions above. Verification: the grid $\mathcal{G}$ consists of 305 distinct multisets (215 uniform, covering every (c, h) with $c \leq 8$, $h(1+c) \leq 120$ or $c \leq 10$, $n \leq 160$; and 90 random mixed multisets with $c_i \leq 12$); at every band position of every member, T1 or T2 passes in exact arithmetic; worst certificate ratio 0.4352.

The separate Band B verifier proves the finite residual left by Section 3.1: after the explicit $C \geq 55$ threshold, it enumerates every leaf-heavy multiset with $C \leq 54$ and nonempty Band B. This is 24,394 multisets and 54,049 coefficient positions, all checked directly in exact integer arithmetic with zero failures. The same script records the large-threshold residuals through $C = 1000$ and their positive boundary minimum at $(C, h) = (55, 35)$; the all- C part is the displayed quotient monotonicity check in Section 3.1.

9.2 Polynomial certification of the E-core

At fixed (k, h) and background pattern $W = E^{h-2-w}(E+2b)^w$ in the uniform case, each bracket of (3) is a polynomial in c of degree at most k (each factor coefficient is a polynomial in c ; the bracket is a finite convolution). The claim “(3) holds for *all integers* $c \geq 1$ ” is therefore decidable exactly: write $R(c) = h[bEW]_k^2 - (h-1)[b^2W]_k[E^2W]_k$, check the leading coefficient is positive, choose B at least the largest real root (any upper bound suffices; we use a numerically located bound, then *prove* positivity beyond it by verifying that all coefficients of the Taylor shift $R(c+B)$ are nonnegative), and evaluate R exactly at every integer in $[1, B]$. The numeric root location plays no role in the rigor; it only sizes the verified range. This certified the 996 instances of Theorem 1.7 with zero failures.

9.3 Audit

Because nearly every verification script was written by the same author as the claim it checks, the full chain was subjected to an adversarial audit: independent re-derivation of every load-bearing identity; an exact six-way sub-bound audit of the charge machinery over 10,751 class instances; independent reconstruction of the certification brackets at random points; and end-to-end brute-force unimodality of every tree in the claimed scope by the original dynamic-programming code. The audit found two implementation-level flaws (an incorrectly indexed ceiling inside dev, masked by unrelated slack, and a finite sentinel acting as a false bound at class-birth positions), both repaired with all headline figures unchanged, and no mathematical flaw. The audit report and all scripts are in the accompanying repository.

The same adversarial methodology was applied to the working conjectures themselves, and twice replaced them. The symmetric form (3), formulated from uniform data, was refuted at lopsided pairs $(1, c, 1^{h-2})$; its exact-allocation weighted replacement was in turn refuted at 1-heavy backgrounds $(c_1, c_2, 1^A)$ — a corner the first sweep could not reach because it capped h at 8 — with smallest witness $S(1, 27, 1^{16})$, $n = 107$, $k = 5$. The surviving hypothesis is the aggregate (2) itself, scanned in exact arithmetic across the flow parameter $t \in \{0, \frac{1}{2}, 1, \frac{3}{2}, 2\}$: 14,410 evaluations over the families $(c, c, 1^A)$, $(1, c, 1^A)$, $(2, c, 1^A)$ ($c \leq 128$, $A \leq 256$), all 200 pairwise-violation witnesses, and 40 random mixed multisets, with zero violations, worst ratio

+0.085, and the margin improving to -5.3 in probes to $A = 1024$. At uniform multisets the aggregate ratio was confirmed to coincide with the symmetric-form ratio (zero mismatches), which is the formal statement that Theorems 1.4, 1.6 and 1.7 certify instances of hypothesis (ii).

The ladder (8) was scanned with the same rigor: 5,298 class-mass profiles in exact arithmetic — uniform multisets ($c \leq 12$, $h \leq 12$), the $(a, b, 1^A)$ families including every pairwise-violation witness ($c \leq 128$, $A \leq 256$), and 60 random mixed multisets — with *zero* rung failures; worst rung ratios 0.917 (uniform), 0.971 $((128, 128, 1^{256}), k = 40)$, 0.884 (random mixed), always at the necessary rung $s = 1$. A Schur analysis over *all* partitions of C into h parts shows the reduction landscape is a threshold phenomenon, not a monotonicity: transfer-by-transfer monotonicity of the rung ratio is *false* (11,278 violations among 49,941 exact Robin Hood steps, margins to -0.23), and in 212 of 1,082 exhaustive (C, h, k, s) cases the maximizing partition is not the most balanced one. A later sweep of the bulk-plus-giant family (a^{h-1}, M) (4,720 instances to $h = 100$, $C = 1200$) found it beating the balanced companion even deep in the tight regime (balanced ratios up to 0.956 beaten), so no domination statement holds; what holds in every observation, without exception, is the ladder itself — the sweep maximum is $0.9865 < 1$, at the minimal gap-two giant $(6^{99}, 8)$, $k = 100$.

Moreover, the polynomial certification extends to the rungs on the two-adjacent-value families $(a^p, (a+1)^q)$ — the target of the Ladder–Schur reduction: at fixed (k, h, p, s) each class mass is a polynomial in a of degree at most $k - s$, and the rung was certified for *all* integers $a \geq 1$ at every instance with $3 \leq h \leq 10$, $1 \leq p \leq h$, $k \leq 28$: 7,866 instances, zero failures (with $p = h$ covering the uniform higher rungs) — the ladder holds for every two-adjacent-value multiset within the reach of the uniform certification grid. As a consistency datapoint, the balanced partition $(1^4, 2^{254})$ of $(C, h) = (512, 258)$ at $k = 40$ has rung-1 ratio 0.9729, slightly above the 0.9711 of the lopsided $(128, 128, 1^{256})$ — the balanced partition is the worse one, as the reduction predicts.

10 Assembly: proofs of Theorems 1.2 and 1.3

Proof of Theorem 1.3. Fix the multiset and let k be a band position (4). Hypothesis (ii) is precisely $\phi'' \leq 0$ on $[0, 2]$ at k , hence the flow facts hold (Proposition 7.1), hence the validity of certificate T2’s constraint rows; certificate T1 is valid unconditionally. By the verified certificates, (6) holds at k ; by Lemma 5.1, P is log-concave at k . Log-concavity at $k = 1$ is universal (Section 3), and band B is covered by Section 3.1. Thus P is log-concave on $[1, k_{\text{dec}} - 1]$ and nonincreasing on $[k_{\text{dec}}, D]$ by Levit–Mandrescu/Basit–Galvin, so Lemma 3.1 applies. Hypothesis (i) enters exactly once, in the penultimate sentence: the master inequality at k is established by the verified certificate at $(c_1, \dots, c_h; k)$. $\square$

Proof of Theorem 1.2. $h = 1$ is Lemma 4.5. For $h = 2$: Theorem 1.4 supplies the symmetric form at every position with constant $2 = 1 + \frac{1}{h-1}$, hence hypothesis (ii) by Proposition 7.2 (a single pair; the AM–GM step), so Theorem 1.3 applies to every $S(c, c)$ in the certificate grid; for the eleven mixed members, hypothesis (ii) is Theorem 1.5 directly (band B is empty at $h = 2$: $k_{\text{dec}} \leq \lceil (2C + 3)/3 \rceil \leq C$ for $C \geq 3$). For the sixteen three-hub adjacent-two-value families, hypothesis (ii) is supplied by the ladder certificates on $(a^p, (a+1)^q)$ ($h = 3$, all integer a , $k \leq 28 \geq k_A$) with Proposition 7.4, hypothesis (i) was verified per multiset (worst certificate ratio 0.38), and band B is empty except at $(1, 1, 2)$, whose single position is checked directly. For $3 \leq h \leq 10$: Theorem 1.7 supplies the symmetric form, over the whole flow box by Lemmas 8.1 and 8.2, at every band position provided k_A does not exceed the certified k -range,

which holds exactly for the table of Theorem 1.2; at uniform multisets the diagonal is symmetric, so Proposition 7.2 yields hypothesis (ii) and Theorem 1.3 applies. $\square$

11 Extremal results and the counterexample search

Definition 11.1 (Balance). For an interior index k of a positive sequence (a_k) , the (local) *balance* is $\min(a_{k-1}, a_{k+1})/a_k$. A positive sequence fails unimodality if and only if some index k satisfies the global condition $\min(\max_{i < k} a_i, \max_{i > k} a_i) > a_k$; local balance > 1 is the width-one special case, and every log-concavity failure observed in trees to date has width exactly one, so local balance is the operative diagnostic.

A valley at k is precisely an interior index with balance > 1 , so searching for a unimodality counterexample is the optimization problem “maximize balance over trees”; every valley bottom is in particular a log-concavity failure. Three structural facts organize the search space.

The offset law. For uniform hub spiders, $I(S(c^h)) = H_c^h + xE_c^h$, and over the range $h \leq 25$, $c \leq 12$ (n up to 256) the unique log-concavity failure sits at offset exactly $h - 2$ below the top degree for every $c \geq 4$, always with width one. The failure is a *junction*: the last index where the fat xE^h hump still contributes against the thin H^h tail. Balance peaks at 0.0439 for $S(4^5)$ ($n = 46$) and decays in both directions.

A high-fragility tree. A beam search over junction trees built from all 2,975 rooted gadgets on at most 11 vertices, seeded with hub-spider multisets, produced a concrete high-fragility witness: a 48-vertex tree (root, four H_4 hub gadgets, one 11-vertex gadget) with a single log-concavity failure of balance 0.04512 — more than double the previous record (0.0208, for the family T^*). It is the first non-log-concave state in the seeded beam log; one 59-vertex extension reaches balance 0.04554, after which the best logged extensions degrade toward disappearance of the dip. The 48-vertex witness has non-monic sequence (leading coefficient 2), which did not occur among the cited Kadrawi–Levit examples.

Why the mechanism self-defends. Writing any tree as root plus rooted gadgets, $P = \prod F_i + x \prod G_i$, a junction valley needs the $\prod F$ side still rising where the $\prod G$ hump ends, while near the junction the coefficients of $\prod F$ organize as elementary symmetric functions e_j of per-gadget weights. Dip existence forces $e_2 < 1$ while the achievable balance behaves like e_3/e_2 , and Newton–Maclaurin gives $e_3/e_2 \lesssim \frac{1}{2}\sqrt{e_2}$: the same log-concavity phenomenon the conjecture asserts is what caps the only known failure mechanism, empirically at balance ~ 0.05 against the required > 1 . Moreover the decreasing-zone theorems protect the junction’s location: the observed failures sit at offset $h - 2$ from the top degree, while the Basit–Galvin decreasing zone covers the top $D - \ell \approx D^2/(D + n)$ positions, and $(h + C)^2 > (h - 2)(3C + h + 1)$ for all $c_i \geq 1$, so every position the known mechanism can reach lies inside the zone where a valley (an ascent $i_k < i_{k+1}$) is already impossible. A unimodality counterexample by this mechanism would need the dip pushed below the decreasing zone, and the balance data above shows the mechanism weakening, not strengthening, in that direction.

Two further observations close the accessible routes: nesting a dipped tree as a gadget under a new root erases its failure (the second-level $F + xG$ smoothing), and products of tree polynomials smooth width-one cliffs away — in exact scans through total size 64, every forest product checked was log-concave outright, and the flat-tail/steep-cliff combination a non-unimodal

product requires never co-occurs in real tree polynomials. All searches and caps are, of course, evidence about accessible scales only; they are recorded here because the structural caps (the offset law, the Maclaurin tension, the decreasing-zone protection) are proof-shaped statements that delimit where any counterexample must live: width ≥ 2 log-convex windows, for which no generator is known.

12 Open problems

Conjecture 12.1 (Flow concavity). *The flow concavity (2) holds for all $h \geq 2$, all $c_1, \dots, c_h \geq 1$, all band positions k , and all $t \in [0, 2]$: the demotion flow $G(t) = [\prod_l (E_l + tb_l)]_k$ is log-concave in t .*

By Theorem 1.3 the conjecture implies unimodality of $I(S)$ for every hub spider. The evidence: Theorems 1.4, 1.5, 1.6 and 1.7 (which prove it at $h = 2$ and at uniform multisets over the certified range), plus the exact aggregate scans of Section 9 (14,410 instances, zero violations, worst ratio 0.085). We emphasize the history recorded in Remark 7.3: two successive pairwise strengthenings of this conjecture were refuted by systematic probing, at lopsided pairs and at 1-heavy backgrounds respectively, while the aggregate form survives both corners with growing margin. The conjecture as stated is exactly what the pipeline uses — nothing weaker suffices for Proposition 7.1, and nothing stronger is true among the natural candidates.

Conjecture 12.2 (All-parameters certificates). *The certificate inequalities T1/T2 of Section 9 hold for every multiset $(c_1, \dots, c_h)$ and every band position, not only on the verified grid. (All certificate ratios are observed monotone or saturating in (h, c, k) , with worst ratio 0.4352.)*

Question 12.3. Does the demotion-flow method extend to root-gadgets of independence number ≥ 3 (the *gadget dichotomy*), and thence to broader classes of trees in the direction of the full Alavi–Malde–Schwenk–Erdős conjecture?

Beyond the two conjectures, three concrete work items would complete the program for all hub spiders. First, Conjecture 12.1 for mixed multisets with $h \geq 3$, for which Proposition 7.4 supplies a t -free route: it suffices to prove the ladder (8). The supporting structure established so far: the rungs are certified for *all* two-adjacent-value multisets $(a^p, (a+1)^q)$ in the range $h \leq 10$, $k \leq 28$ (all integer a , Section 9), and the exact scans show a threshold landscape — whenever a lopsided partition beats the balanced one, all ratios in that case sit below 0.544, while the near-1 corners are dominated by balanced partitions. Three structural shortcuts are provably unavailable: plain Schur reduction (transfer monotonicity toward balance fails), its conditional form (violating transfers exist at ratio 0.868), and a value-window bound (one outlier in a large balanced bulk leaves the ratio at 0.944). What survives every exact probe ($\sim 25,000$ cases) is endpoint domination in its canonical conditioning: whenever the *balanced companion's* rung ratio is at least 0.7, no multiset with the same (C, h) beats it — the balanced companion is never beaten above ratio 0.66 in any scan, while at low ratios beats are common and harmless. This *domination lemma* is the open core. Two pieces of supporting structure are in place: Proposition 7.5 (both sides of every rung grow away from balance, so domination is a relative-rate statement), and the *band-bottom* reduction: by Proposition 7.5, $\rho_s(c, k) \leq \bar{\rho}(C, h, k, s) := (s+1)W_{s-1}(\text{ext})W_{s+1}(\text{ext})/(sW_s(\text{bal})^2)$ with $\text{ext} = (1^{h-1}, C-h+1)$, an explicit inequality between two canonical multisets. Exact maps ($h \leq 60$, $C \leq 2000$) give $\bar{\rho} \leq 0.546$ at $k - s = 1$ (tracking the derived asymptote $(2 - \frac{3}{2}\lambda)/(2 - \lambda)^2 \leq \frac{9}{16}$, $\lambda = (s-1)/h$) and $\bar{\rho} \leq 0.897$

at $k - s = 2$, while $\bar{\rho}$ crosses 1 at $k - s = 3$ — the elementary reduction reaches exactly to where the high regime begins, so the open domination lemma is confined to $k - s \geq 3$. At rung 1 the ratio is the quadratic form $\sum_{i \neq j} w_i w_j K(c_i, c_j)$ in the normalized weights $w_i = x_i / \sum x$, with the h -free kernel $K(\sigma, \tau) = g y(\sigma + \tau) / (x(\sigma) x(\tau))$, which is log-submodular (equivalently, y is log-concave in its argument) and maximized at extreme pairs that real multisets cannot weight — the mechanism of domination is this weight–kernel anticorrelation, and exact probes rule out every reduction that decouples a pair from the rest of the multiset. What the probes leave standing is a variational program: over the transfer graph at fixed (C, h, k) , all observed tight local maximizers are either adjacent-two-value or of the *bulk-plus-giant* form $(a, \dots, a, M)$. No domination between the families survives scrutiny — exact sweeps of the giant family show it beating the balanced companion deep in the tight regime (e.g. $(1^5, 95)$ at $k = 46$: 0.778 against 0.734; balanced ratios up to 0.956 are beaten), while the ladder itself stays unviolated, with the sweep maximum 0.9865 attained at the minimal gap-two giant $(6^{99}, 8)$. Domination is also unnecessary: if every tight local maximizer is one of the two shapes, the rung for all multisets follows from per-family bounds alone. Both candidate-maximizer families are certified within the standing caps: the adjacent-two-value rungs (7,866 instances) and the bulk-plus-giant rungs (1,512 instances, $h \leq 10$, $M - a \leq 8$, $k \leq 28$, all integer a , zero failures). An exact hill-climbing census at scale ($h \leq 24$, $C \leq 128$, ~ 18 starts per position) found 76 distinct local maximizers, every one in the two families. The open core is therefore the first-order *exchange lemma*: every tight local maximizer of a rung over the transfer graph is adjacent-two-value or bulk-plus-giant. Its proof must treat the stationarity system globally — the giant’s stability is collective (the pair (a, M) is stable only because the bulk pins the background), the structural reason the pairwise reductions above fail. Second, monotonicity or limit arguments upgrading the certificate grid to all parameters (Conjecture 12.2). Third, an aggregate analysis of the symmetric form in the deep band of the uniform case, where the defect is observed to be negative but termwise and low-order symmetrization arguments provably fail; the higher ladder rungs, which carry more slack than the E-core rung in all scans, may help here too.

AI-use Declaration

The author used AI tools, including Claude (Anthropic) and ChatGPT/Codex (OpenAI), during proof exploration, drafting, code development, audit, and reproducibility packaging. These tools are not authors. The human author reviewed the output, takes responsibility for the manuscript, proofs, verification code and data, and approved the final version. All verification scripts and paper-facing logs are available in the accompanying repository.

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